Magnetic Polarity Detector Using Hall Effect

Abstract & Core theory

This project presents a Magnetic Polarity Detector using a Hall Effect Sensor.
It detects North and South magnetic poles without physical contact and displays the result using LED indicators.
Core Theory (Hall Effect)
When current flows through a conductor and a magnetic field is applied, charge carriers shift, creating a Hall Voltage.
The direction of this voltage determines the magnetic polarity.

component

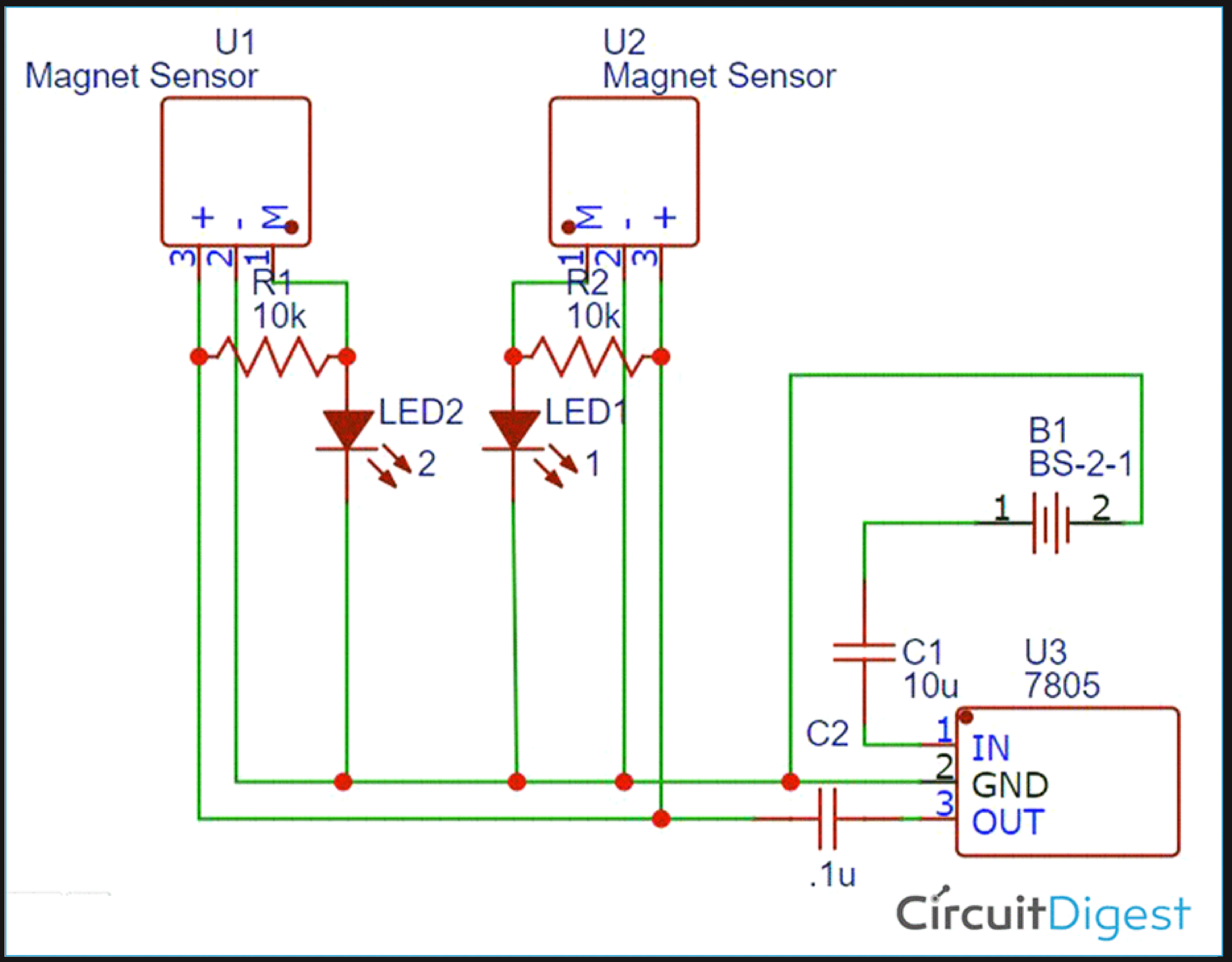
- Hall Effect Sensor
- 9V DC Power Supply
- 7805 Voltage Regulator (5V)
- Resistors (Pull-up & LED protection)
- LEDs (Red & Yellow)
- Breadboard & Jumper Wires

flow

Magnet → Hall Sensor → Signal Processing → LED Output

Implementation challenges

- Incorrect sensor orientation
- Weak magnetic field
- Electrical noise
- Unstable power supply
- Long connecting wires



Supervisors

Dr: Nur ad-Din M Salem

Eng: Ehab Ahmed

Team members

- Mostafa Hatem.
- Yousef Abdallah.
- Mennat Elrahman.
- Mriam Mansour